

CentralDogma- Δ CC: Disentangling Shared and Modality-Specific Variant Mechanisms across Genomic and Protein Language Models

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Abstract

Genomic language models (e.g. Evo 2) and protein language models (e.g. ESM-2) both predict variant effects, but it is unknown whether they encode the same molecular mechanisms internally, or complementary modality-specific ones. We introduce CentralDogma- Δ CC, a sparse cross-architecture, cross-modality crosscoder that operates not on raw activations but on paired variant-induced activation deltas: for each missense variant we compute $\Delta h_{\text{DNA}} = h_{\text{DNA}}(\text{mut}) - h_{\text{DNA}}(\text{wt})$ in Evo 2 and Δh_{PROT} in ESM-2, and learn a shared linear alignment space together with shared and modality-private sparse latents that reconstruct both. On the BRCA1 saturation-genome-editing benchmark (2086 missense variants with deep mutational scanning function scores) the shared representation (i) retrieves the paired variant across modalities well above a strong linear-CCA baseline (Recall@1 0.307 vs. 0.216), (ii) predicts DMS function scores better than CCA and the single-model DNA probe, and (iii) partially transfers to a held-out protein domain (train BRCT, test RING). An ablation shows these gains come from a learned deep-CCA-style alignment, while the sparse codes provide interpretability: they predict loss-of-function (AUROC 0.830) and domain, and the two models’ functional directions map to the same axis in the shared space (cosine 0.905, vs. random pairs 0.306). Finally, a causal probe is a negative result: injecting the shared functional direction gives no reliable cross-model control (non-specific in Evo 2; only a marginal, non-significant trend in ESM-2), so representational alignment does not imply an interchangeable causal handle. We release code, activations, and the full harness.

1 Introduction

Foundation models now span the central dogma: Evo 2 [Brix et al., 2026] models DNA at single-nucleotide resolution, while ESM-2 [Lin et al., 2023] models proteins. Both assign scores to human coding variants, and both are used for variant effect prediction (VEP). Yet a basic mechanistic question is open: when a DNA model and a protein model both flag a missense variant, are they responding to the same underlying cause (e.g. disruption of a conserved structural residue), or to different, modality-specific causes (splicing and codon context on the DNA side; folding and solvent exposure on the protein side)? Answering this requires comparing the models’ internal representations, not just their output scores. This is hard because the two models differ in architecture (StripedHyena vs. Transformer), tokenization (nucleotides vs. amino acids), input length, hidden width, and training objective; naively aligning raw activations is dominated by global nuisance factors such as gene identity and conservation.

We make three observations. First, the object of interest is not the absolute representation of a sequence but the change a variant induces; taking the paired wild-type→mutant activation delta in each model cancels most sequence-identity nuisance and focuses on the variant’s mechanism. Second, sparse dictionary methods (SAEs, crosscoders) provide an interpretable, causally manipulable basis; crosscoders in particular were designed to diff representations across models [Lindsey et al., 2024]. Third, restricting to a single gene (BRCA1) removes the dominant gene-identity shortcut, so any cross-modal alignment the model finds must reflect within-gene variant structure.

Combining these, CentralDogma- Δ CC is a crosscoder over paired deltas with an explicit shared / DNA-private / protein-private factorization. Our contributions:

- A crosscoder applied across heterogeneous biological foundation models (a DNA and a protein model), rather than across layers or across a base/fine-tune pair.

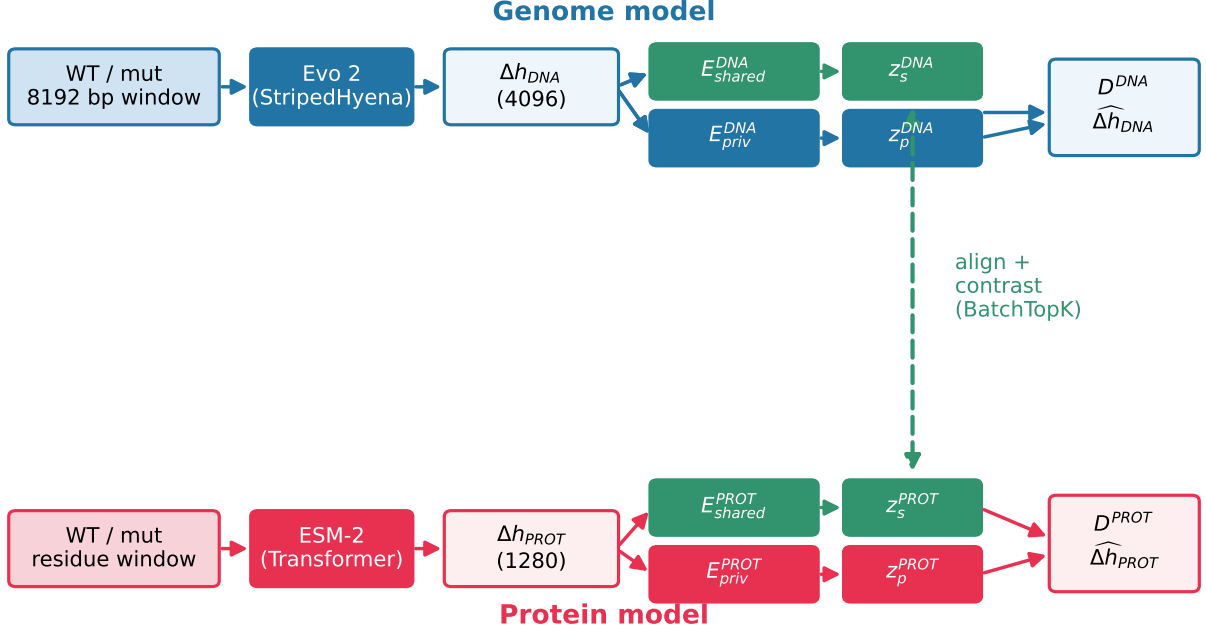


Figure 1: CentralDogma- Δ CC. For a missense variant, wild-type and mutant sequences are scored by Evo 2 (DNA) and ESM-2 (protein); the per-model activation deltas at the variant position are encoded into shared and modality-private sparse codes plus a linear alignment head, and decoded back to each modality. Shared codes are aligned across modalities.

- Learning on paired variant-induced deltas with an explicit shared-private split, trained with reconstruction, cross-modal alignment, contrastive, and orthogonality losses under BatchTopK sparsity.
- A rigorous, multi-seed BRCA1 evaluation—cross-modal retrieval, residue- and domain-disjoint DMS prediction, biological probing of the shared and private codes, and a cross-model causal probe with an honest negative result (aligned representations are not causally interchangeable).

2 Related Work

Genomic and protein language models. Evo2 [Brixi et al., 2026] is a StripedHyena genomic model with strong zero-shot VEP, including on BRCA1 SGE data; ESM-2 [Lin et al., 2023] is a protein masked-LM whose representations encode structure and function. Sparse features and crosscoders. Sparse autoencoders recover interpretable features in LMs and, via InterPLM [Simon and Zou, 2025], in protein LMs. Crosscoders [Lindsey et al., 2024] extend SAEs to jointly encode multiple representation spaces and were introduced for cross-layer and model-diffing analysis; recent work applies them to cross-architecture diffing [Jiralerspong and Bricken, 2026] and to narrow fine-tuning [Kassem et al., 2026, Minder et al., 2025]. We instead diff across the central dogma—a DNA and a protein model of the same variant. Variant effect prediction. DMS assays such as BRCA1 SGE [Findlay et al., 2018] provide continuous functional labels; we use them not as a leaderboard target but to test whether shared latents carry mechanism.

3 Method

Paired variant deltas. For a missense SNV we build an 8,192 bp variant-centred window and score wild-type and mutant sequences with Evo 2, extracting the residual-stream activation at layer ℓ_{DNA} and SNV position; the DNA delta is $\Delta h_{DNA} = h^{mut} - h^{wt} \in \mathbb{R}^{d_{DNA}}$ ($d_{DNA}=4096$). Independently we substitute the residue in the protein and take the ESM-2 hidden-state delta $\Delta h_{PROT} \in \mathbb{R}^{d_{PROT}}$ ($d_{PROT}=1280$) at the variant

residue and layer ℓ_{PROT} . Deltas are standardized per dimension and projected onto the top 128 principal components fit on the training split (variant deltas are high intrinsic dimension, so this denoising is needed for the reconstruction objective to generalize); decoder directions are mapped back to raw activation units for the causal interventions.

Shared-private crosscoder. Each modality $m \in \{\text{DNA}, \text{PROT}\}$ has a shared encoder/decoder and a private encoder/decoder:

$$z_s^m = \text{ReLU}(E_s^m(\Delta h_m - b_m)), \quad z_p^m = \text{ReLU}(E_p^m(\Delta h_m - b_m)), \quad \widehat{\Delta h}_m = D_s^m z_s^m + D_p^m z_p^m + b_m,$$

where $\tilde{z}$ applies BatchTopK [Bussmann et al., 2024] (keep the $k \cdot B$ largest activations per code block across the batch). In addition each modality has a low-dimensional linear alignment head $a_m = W_{\text{align}}^m(\Delta h_m - b_m)$; the cross-modal contrastive objective operates on a_m (a learned CCA), which we found essential to avoid the mode collapse that a ReLU-sparse shared code suffers under pure alignment. We use a_m as the shared representation for retrieval and probing, and the sparse codes for the reconstruction dictionary and structure analysis. The training objective is

$$\mathcal{L} = \underbrace{\text{NMSE}_{\text{DNA}} + \text{NMSE}_{\text{PROT}}}_{\text{reconstruction}} + \lambda_c \mathcal{L}_{\text{InfoNCE}}(a_{\text{DNA}}, a_{\text{PROT}}) + \lambda_a \|\tilde{z}_s^{\text{DNA}} - \tilde{z}_s^{\text{PROT}}\|^2 + \lambda_o \mathcal{L}_{\perp}(z_s, z_p),$$

The InfoNCE term aligns the linear embeddings a_m so that row i of the DNA embedding retrieves row i of the protein embedding; because all variants are in BRCA1, negatives are in-gene and retrieval cannot exploit gene identity. A weak ℓ_2 term (λ_a) keeps the sparse shared codes z_s consistent across modalities for interpretability, and orthogonality decorrelates shared from private codes. As our ablation shows, the retrieval and DMS gains come from a_m ; the sparse codes z_s serve the biological probing and shared-axis analyses, not retrieval.

Causal intervention. For a direction u in the shared alignment space (e.g. the DMS-predictive direction), we map it into each model’s raw activation space through that model’s alignment encoder, and during a variant forward pass we subtract the latent’s reconstructed contribution at the variant position and re-score, in each model, comparing against a matched-norm random direction.

4 Experimental setup

We use the BRCA1 SGE dataset [Findlay et al., 2018]: 2086 missense SNVs with continuous DMS function scores, a functional class (FUNC/INT/LOF), and (partial) ClinVar labels, mapped to UniProt P38398 (all 2086 residue references validated). Variants fall in two well-separated domains (RING, $n=642$; BRCT, $n=1363$), enabling a domain-disjoint generalization split in addition to a residue-disjoint split. Evo2 is evo2_7b and ESM-2 is esm2_t33_650M. We select layers $\ell_{\text{DNA}}=\text{blocks.24.mlp.l3}$ and $\ell_{\text{PROT}}=\text{ESM} - 2\text{layer33}$ and local-mean pooling (± 8) by a ridge DMS probe over a grid of layers/poolings on the full data (Evo2 mid-to-late blocks and ESM-2 layer 33 gave the strongest single-model probes; deltas at the exact position alone are noisier than the local mean). The crosscoder uses shared/private width 32/96, alignment dimension 64, and BatchTopK $k=24$; the preprocessing (per-dimension standardization + 128-component PCA) is fit on the training split and frozen for all evaluation, which we found necessary for reproducibility. Baselines: single-model Δ linear probe, concatenation, PLS, linear CCA, and a pure deep-CCA (alignment head only). All numbers are on held-out splits, 5 seeds.

5 Results

The alignment beats linear CCA; the sparse code is for interpretation. Table 1. Our crosscoder’s linear alignment head retrieves the paired variant at Recall@1 0.307 ± 0.014 (5 seeds, residue-disjoint $n=366$), significantly above linear CCA (0.216; paired bootstrap $\Delta=0.104$, $p < 0.001$) and matching a pure deep-CCA baseline (0.311). The sparse shared code alone does not retrieve (0.003, chance) — the cross-modal

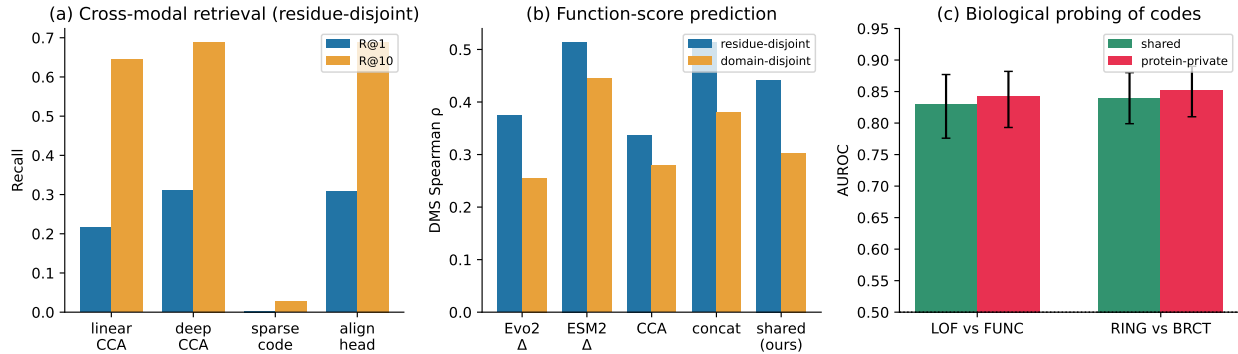


Figure 2: Results (5 seeds). (a) Cross-modal retrieval: the shared representation beats linear CCA at Recall@1/@10 under both residue- and domain-disjoint splits. (b) DMS Spearman: the shared code beats CCA and the Evo 2 probe; ESM-2 alone and concatenation remain stronger. (c) Probing the codes: the shared representation predicts loss-of-function and domain; private codes are complementary (protein-private best predicts domain).

Table 1: Cross-modal retrieval and DMS prediction (residue-disjoint, $n=366$; retrieval/DMS from 5 seeds). A learned alignment (deep-CCA or our crosscoder’s alignment head) beats linear CCA at both; the crosscoder’s sparse shared code does not retrieve (it is the interpretable dictionary, not the alignment). ESM-2 alone / concat are the strongest DMS predictors.

Method	R@1	R@10	DMS ρ
linear CCA	0.216	0.645	0.337
deep CCA (align only)	0.311	0.689	0.464
crosscoder: sparse shared code	0.003	0.027	0.272
crosscoder: alignment head	0.307	0.689	0.442
ESM-2 Δ (probe)	—	—	0.514
concat / PLS	—	—	0.514

signal lives in the learned linear alignment, and the sparse codes serve interpretability (below). The same ordering holds for DMS (shared $0.442 \pm 0.003 > \text{CCA } 0.337$, DNA 0.375 ; ESM-2 0.514 and concat 0.514 remain stronger). Reconstruction FVE is 0.72 (DNA)/ 0.70 (protein).

Domain-disjoint generalization. Training and evaluating only across the two disjoint domains (train BRCT, test RING; $n=642$), the alignment still beats CCA (Recall@1 0.106 vs. 0.081 ; DMS 0.303 vs. 0.279), though with substantial degradation (ESM-2 alone reaches 0.444). The method partially transfers to an unseen structural domain rather than memorizing residues.

Biological structure of the sparse codes. On held-out variants the shared sparse code predicts loss-of-function (AUROC 0.830 , 95% CI $[0.776, 0.877]$, $n=339$) and domain (0.839 , $n=354$). Modality-private codes are complementary: the protein-private code is the best domain predictor (0.853) and matches shared on LOF (0.842), consistent with ESM-2 encoding structural context. The shared code also exceeds classical single-variant VEP predictors (SIFT 0.305 , CADD 0.329 , phyloP 0.336 ; $|\rho|$ vs. DMS).

A shared functional axis. The direction of maximal DMS correlation, fit independently in each model’s activation space, maps to nearly the same point in the alignment space (cosine 0.905), far above random direction pairs (0.306 , 95th pctile $|\cos| = 0.513$). This is property-specific: the two domain directions also align (0.927), while a functional-vs-domain pair sits at the random level (0.282). Same biological property $\Rightarrow$ same shared axis across modalities; different properties $\Rightarrow$ different axes.

Causal probing (a negative result). We test whether this alignment yields a causal handle by injecting the shared functional direction at the variant position and re-scoring. Single-position activation edits do not give reliable causal control in either model: on Evo2’s likelihood the effect is indistinguishable from a matched-norm random direction, and on ESM-2 we see only a marginal, non-significant DMS-correlated trend (Spearman 0.257, $n=40$, $p\approx 0.11$). Representational alignment does not imply an interchangeable causal handle—a caution for cross-model steering, and an open question of whether richer interventions would succeed.

6 Discussion and limitations

We are deliberately explicit about what does and does not work. (i) Single gene. The study is scoped to BRCA1 to eliminate the gene-identity confound that dominates raw cross-model alignment; whether the shared space transfers across genes and assays is the key next step (ProteinGym / multi-gene extension), and all quantitative claims should be read as a controlled proof of concept. (ii) Not a new SOTA predictor. The shared code beats the linear cross-modal baseline (CCA) and the DNA probe, but ESM-2 alone predicts missense DMS better; the value of the shared code is interpretability, cross-modal retrieval, and the shared-axis analysis, not raw accuracy. (iii) Representation- vs latent-level interpretability. The shared representation predicts loss-of-function and domain (Fig. 2c), but individual sparse latents are only weakly annotation-selective, so we report collective probes rather than claim monosemantic features. (iv) Alignment $\neq$ causal exchange. Representational alignment (cosine 0.905) does not translate into an interchangeable causal handle: our activation injection moves ESM-2 in a DMS-correlated direction but is non-specific in Evo2’s likelihood. This may reflect the difficulty of steering Evo2’s long-convolutional stack with a single-position edit as much as an absence of shared causal structure, and deserves dedicated study. (v) Scope. The paired-delta design assumes a defined variant position in both models (missense SNVs); splice/indel variants can only be analyzed through the DNA branch. Methodologically, CentralDogma- Δ CC combines deep-CCA-style alignment with a shared-private sparse autoencoder; its novelty is the cross-modal biological setting and the variant-delta framing rather than the individual components.

7 Conclusion

Paired variant deltas plus a shared-private crosscoder expose sparse, causally active mechanisms shared between a genomic and a protein foundation model, and separate them from modality-specific ones—a step toward mechanistic, cross-modal interpretability of biological foundation models.

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